

# When is a topological helicity current a matter current? A no-go trilogy and a conditional regularity closure for driven force-free plasmoids

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**Abstract.** For a force-free (Beltrami) field  $\nabla \times \mathbf{B} = \lambda \mathbf{B}$  and its two-fluid generalization we ask when the fluid mass 4-current  $(\rho, \rho \mathbf{v})$  may be identified with the topological helicity 4-current  $(K^0, K^i)$ ,  $K^0 = \mathbf{A} \cdot \mathbf{B}$ . The density leg  $\rho \propto \mathbf{A} \cdot \mathbf{B}$  holds exactly as a Beltrami-gauge profile identity. The *current* leg is settled as a tiered result: **(i)** in the static regime the 4-current closes iff  $|\mathbf{B}| = \text{const}$ , which a topologically nontrivial (nulled) field cannot satisfy — a proven no-go; **(ii)** the two-fluid escape via a conserved *canonical* helicity only relocates the obstruction to  $\mathbf{P} \cdot \mathbf{v} = \text{const}$ , again impossible for a nontrivial field via a perfect-square identity; **(iii)** in the genuinely driven regime closure reduces to a single scalar condition  $S = \mu - \mathbf{P} \cdot \mathbf{v} = 0$ , whose surface is non-empty and realizable by an electrostatic drive, while the equilibrium and relaxation dynamics *select against* it — closure is externally driven, never emergent. The one remaining gate — a global enstrophy / Beale–Kato–Majda bound — we close **conditionally**: an exact identity recasts the vortex-stretching integral as a Lamb-vector flux, converting the supercritical enstrophy inequality into a *linear* one and yielding an a-priori bound on the driven near-Beltrami limit cycle provided the time-averaged Beltrami deviation stays below an explicit  $O(1/\text{Re})$  threshold. The result lifts to resistive–viscous Hall two-fluid at unit magnetic Prandtl number, with an explicit  $(\eta - \nu)^2$  obstruction otherwise. Tiers are stated throughout ([V] proven here; [credited] established); no over-unity is claimed.

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## 1. Introduction

Force-free fields  $\nabla \times \mathbf{B} = \lambda \mathbf{B}$  (Chandrasekhar & Kendall 1957; Woltjer 1958) are the minimum-energy states of magnetically dominated plasmas at fixed magnetic helicity  $H = \int \mathbf{A} \cdot \mathbf{B}$  (Taylor 1974; Moffatt 1969). A recurrent and physically attractive idea — in models where matter and field co-move — is that the fluid mass current *is* the flux of the field’s own winding: that  $\rho$  measures how densely the field is wound and  $\rho \mathbf{v}$  measures how that winding flows. This paper asks precisely when that identification is

mathematically admissible, treating the fluid current and the helicity current as candidate equal 4-vectors and testing the equality leg by leg.

The answer is a trilogy — two proven no-gos (static; aligned/steady), one constructive realizability result (driven), and a first-class selection negative — plus a clean reduction of the last gate to a recognized regularity problem, which we then close under an explicit, falsifiable hypothesis on the drive. Throughout, the scope discipline is that these are limits of a *construction*, not verdicts against any observed plasma phenomenon.

## 2. The law, its two legs, and the density leg [V]

Write ordinary matter conservation as a candidate topological continuity equation,  $\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0 \Leftrightarrow \partial_\mu K^\mu = 0$ , with  $K^\mu = (K^0, K^i)$ ,  $K^0 = \mathbf{A} \cdot \mathbf{B}$  the helicity density and  $K^i$  its flux, under the organizing proportionality  $\rho = \kappa(\mathbf{A} \cdot \mathbf{B})$ . Two legs must hold for the 4-current to close: the **density leg**  $\rho \sim K^0$  and the **current leg**  $\rho \mathbf{v} \sim K^i$ .

In the Beltrami gauge  $\mathbf{A} = \mathbf{B}/\lambda$ ,  $K^0 = \mathbf{A} \cdot \mathbf{B} = |\mathbf{B}|^2/\lambda \geq 0$  (sign-definite, verified to  $3 \times 10^{-15}$ ). With a London-locked cold-condensate flow  $\nabla \times \mathbf{v} = -(q/m)\mathbf{B}$  (verified to  $7 \times 10^{-15}$ ), the density profile tracks  $|\mathbf{B}|^2$  with  $\text{corr}(\rho, \mathbf{A} \cdot \mathbf{B}) = 1.0000$  — an exact dimensionless profile identity;  $\kappa$  is a genuine dimensional constant. **The density leg is settled**; the entire subtlety lives in the current leg.

## 3. The static no-go: closure requires $|\mathbf{B}| = \text{const}$ [V]

Matching densities fixes the current *direction*: London flow gives  $\mathbf{v} \sim \mathbf{A} \sim \mathbf{B}$ , so the matter current  $\rho \mathbf{v} \sim |\mathbf{B}|^2 \mathbf{B}$  is **cubic** in the field while the helicity flux  $K \sim \mathbf{B}$  is **linear**. Equality of a cubic and a linear quantity demands  $|\mathbf{B}|^2 = \text{const}$ :

*The 4-current closes if and only if  $|\mathbf{B}|$  is constant.*

The obstruction is basis-independent, measured by the residual of  $\mathbf{B}$  from  $\text{range}(\mathbf{L})$  for  $\mathbf{L}[\phi] = \nabla \times ((\phi \mathbf{B} - \nabla \phi \times \mathbf{B})/|\mathbf{B}|^2)$  — dimensionless and  $\kappa$ -independent. For a genuine toroidal Beltrami field the residual is 0.6053 (basis-converged); a four-move reducibility sweep (gauge, transverse  $\mathbf{E}$ /sheath, norm, multi- $\lambda$ ) never reaches the  $< 0.2$  bar, and a transverse non-force-free  $\mathbf{E}$ -field leaves it unchanged. The deciding control is the null-free constant- $|\mathbf{B}|$  Beltrami field  $\mathbf{B} = B_0(\cos z, \sin z, 0)$ , which closes exactly (residual 0.0000). A topologically nontrivial closed-fibre field must have nulls

and therefore cannot have  $|B| = \text{const}$ . **In the static regime the current leg is over-determined by construction — a postulate, not a theorem.**

#### 4. Canonical helicity, and the aligned/steady no-go [V]

A driven object may carry its own conserved current. For a two-fluid plasma the correct one is **canonical (generalized) helicity**: with  $P = A + (m/q)v$  and  $\Omega = \nabla \times P = B + (m/q)\nabla \times v$ , the current from  $H = \int P \cdot \Omega$  is ideally conserved even when magnetic helicity is dissipated, because the canonical Ohm's law  $E_{\text{can}} = -v \times \Omega$  gives  $E_{\text{can}} \cdot \Omega = 0$  (a Casimir invariant) (Steinhauer & Ishida 1997; Mahajan & Yoshida 1998).

Having the conserved current only *relocates* the obstruction. The canonical flux is  $K_{\text{can}} = h v + (\mu - P \cdot v) \Omega$  ( $h$  the canonical helicity density,  $\mu$  the Bernoulli head), so closure requires  $(\mu - P \cdot v) \Omega = 0$ . On a double-Beltrami equilibrium  $\mu = \text{const}$ , and this is the exact mirror of the static theorem:

*Aligned/steady closure requires  $P \cdot v = \text{const}$  (the analog of  $|B| = \text{const}$ ).*

Making the relevant cross-term vanish forces  $\lambda_- = -\lambda_+$ ; then  $P \cdot v = \text{const}$  requires a positive ratio  $p_1^2/p_2^2 > 0$  between two helicity-fluctuation triples, which is impossible because  $(A'B')(B'C')(A'C') = (A'B'C')^2 \geq 0$  is a perfect square. Independent checks confirm it: the residual reaches zero only at the forbidden-sign branch  $c = -0.604 < 0$ ; distinct  $\lambda$  force a spectral band-gap and single-mode collapse; and the field-magnitude inhomogeneity equals  $\text{std}(P \cdot v)/\text{mean}$  to  $\sim 1\%$ . The two no-gos are thereby unified — the same obstruction seen once in  $B$  and once in  $P$ .

#### 5. The driven resolution, and the selection negative [S→V] / [V]

The third leg differs in kind. Define the scalar  $S = \mu - P \cdot v$ . Since  $S$  is a scalar,  $S\Omega = 0$  forces  $S = 0$  pointwise (no “ $S \perp \Omega$ ” branch; verified false to  $4 \times 10^{-17}$ ). The canonical current is then always a matter current  $(h, h u)$ ,  $u = v + (S/h)\Omega$ , closing on  $v$  exactly when  $S = 0$ . Two facts make this a genuine resolution. **(a)** The  $S = 0$  surface is non-empty and drive-compatible (jointly satisfiable to  $\sim 10^{-16}$ ), unlike the two no-gos whose surfaces are empty; a degree-of-freedom count is sharp ( $8/8$  isolated  $\rightarrow 9/8$  over-determined for closure  $\rightarrow$  the spare freedom supplied by a sustained drive). **(b)** No extra  $f_{\text{ext}} \perp \Omega$  constraint is needed: the exact identity  $\partial_\mu K_{\text{can}}^\mu = 2 f_{\text{body}} \cdot \Omega$  vanishes for a purely electromagnetic (Lorentz + barotropic) force, so the drive is simply the electrostatic potential  $\phi = A \cdot v + (m/2q)|v|^2 - w/q$ ; the differential-algebraic system is index-1 ( $dS/d\phi = 1 \neq 0$ ), so  $\phi$  is

algebraically slaved and bounded, with two-species Jacobian  $\det = -|v_e - v_i|^2$  (invertible iff current flows). **Tier: [S] leaning [V]**.

Does the object select  $S = 0$  on its own? **No** — a tested negative. *Variationally*, minimizing energy at fixed canonical helicities yields double-Beltrami equilibria with  $\mu = \text{const}$ , hence  $S = \text{const} - P \cdot v \neq 0$ . *Dynamically*, a driven-dissipative Hall two-fluid integration keeps  $\text{rms}(S)/|P \cdot v|$  at  $O(1)$  and relaxes *toward* the aligned  $\mu = \text{const}$  state. **Closure is externally driven, never emergent** — and, as a corollary, the two no-gos are the dynamical attractor of relaxation, reinforcing them.

## 6. The regularity gate, closed conditionally [V] conditional

On a bounded, current-full trajectory the remaining residual reduces to a global existence question: given an energy/ $L^2$  bound (a Lyapunov function  $V = (r^2 - R_0)^2/4$ ,  $dV/dt \leq 0$ , global attraction to the saturated heartbeat amplitude  $r^* = \sqrt{2}$ ) and no current-null crossing (a measure-zero, benign set), does the driven near-Beltrami limit cycle admit an a-priori **enstrophy / Beale-Kato-Majda** bound  $\int_0^\infty \|w\|_\infty dt < \infty$ ? This is category-identical to open large-data 3D regularity (Beale, Kato & Majda 1984; Chae, Degond & Liu 2014). We close it **conditionally**, without solving general Navier–Stokes.

**Lemma (exact vortex-stretching = Lamb-vector flux).** For a smooth divergence-free field  $v$  on  $\mathbb{T}^3$ ,  $w = \nabla \times v$ ,

$$P := \int w \cdot (w \cdot \nabla) v \, dx = \int (\nabla \times w) \cdot (v \times w) \, dx. \quad (1)$$

*Proof.* The identity  $(w \cdot \nabla) v = \nabla \times (v \times w) + (v \cdot \nabla) w$  (both fields divergence-free) and  $\int w \cdot (v \cdot \nabla) w = \frac{1}{2} \int v \cdot \nabla |w|^2 = 0$  give  $P = \int w \cdot \nabla \times (v \times w)$ ; integrating by parts yields (1). ■ Thus vortex stretching is *exactly* a flux of the **Lamb vector**  $L = v \times w$ , and vanishes identically at a force-free state ( $L = 0$ ) — production is controlled *linearly* by the deviation, not by a generic strain norm. Both forms of (1) agree to ~9 significant figures numerically on a generic field where  $P \neq 0$ .

**Linear inequality.** The enstrophy balance  $dZ/dt = P - \nu \|\nabla w\|_2^2 + F$  ( $Z = \frac{1}{2} \|w\|_2^2$ ) with (1), Cauchy–Schwarz  $|P| \leq \|\nabla w\|_2 \|L\|_2$ , Young, and Poincaré  $\|\nabla w\|_2^2 \geq \lambda_1 \|w\|_2^2$  gives

$$dZ/dt \leq -\beta(t) Z + F(t), \quad \beta(t) = \nu \lambda_1 - \eta(t)^2/\nu, \quad (2)$$

where  $\eta = \delta \|v\|_\infty$  and  $\delta = \|v \times w\|_2 / (\|v\|_\infty \|w\|_2) \in [0, 1]$  is the Beltrami deviation. The classical route yields instead a supercritical  $Z^3$  term; the exact identity (1) is what makes (2) **linear**.

**Theorem (conditional a-priori enstrophy bound).** On the  $T$ -periodic driven orbit, if the amplitude is bounded ( $\|v\|_\infty \leq U^*$ , the coherent-object regime) and the time-averaged deviation is sub-threshold,

$$\langle \eta^2 \rangle := (1/T) \int_0^T \eta^2 dt < v^2 \lambda_1 \quad \Leftrightarrow \quad \delta_{\text{rms}} \lesssim 2\pi/\text{Re}, \quad \text{Re} = U^*L/v, \quad (3)$$

then  $\sup_t Z(t) < \infty$ . Consequently  $\|w\|_2$  is uniformly bounded; by the 3D  $H^1 \Rightarrow$ regularity criterion the orbit is globally smooth and the Beale–Kato–Majda integral stays finite — **the driven heartbeat is globally regular**. (*Proof: forcing is absorbed by Young;  $\langle \mathcal{B} \rangle > 0$  gives a Gronwall bound  $\sup Z \leq e^{2\tilde{B}T}(Z(0) + 2Q/\langle \mathcal{B} \rangle)$ .) Condition (3) is **time-integrated**, not pointwise: the deviation may spike each heartbeat provided its average stays sub-threshold — verified numerically, with the boundedness transition sharp at  $\langle \eta^2 \rangle = v^2 \lambda_1$ .*

**Hall two-fluid lift.** The ion generalized vorticity  $\Omega = B + d_i w$  is frozen into the ion flow,  $\partial_t \Omega = \nabla \times (v \times \Omega) + \text{dissipation}$ , so (1) ports verbatim to the *independent* divergence-free pair  $(v, \Omega)$  (verified to machine precision), and the canonical Lamb vector vanishes on the double-Beltrami state. The conditional bound carries over at **unit magnetic Prandtl number**  $\text{Pm} = \nu/\eta = 1$ , where the two-fluid dissipation collapses to the perfect square  $\eta \|\nabla \Omega\|_2^2$ ; for  $\text{Pm} \neq 1$  the dissipation quadratic form is indefinite ( $\det = -d_i^2(\eta - \nu)^2/4 < 0$ ), the precise obstruction that localizes the remaining Hall difficulty.

## 7. Scope and discussion

The identification “matter is the flow of the knot” is admissible only for a knot that is being actively driven; in a frozen or self-organizing field it is provably barred. The lead statement is a **[V] no-go / well-founded postulate in the frozen regimes** (static and aligned/steady, both proven), a **constructed-realizable closure  $[S \rightarrow V]$  in the driven regime**, a **first-class selection negative** (it is driven, not emergent), and a **conditional [V] closure of the regularity gate** under the explicit drive hypothesis (3). What is *not* claimed: unconditional closure — at the object’s Lundquist number  $S \sim 10^2\text{--}10^6$  the  $O(1/\text{Re})$  condition (3) is stringent and unproven, and the  $\text{Pm} \neq 1$  Hall case requires a coupled fluid+magnetic enstrophy functional. No over-unity is claimed; the density leg stays [V]; canonical helicity is a genuine ideal conserved current  $[V]/[\text{credited}]$ . The results stand independently of any wider synthesis: a self-contained plasma / topological-fluid contribution.

## 8. Conclusion

Whether a topological helicity current can be a matter current is decided leg by leg: the density leg holds; the static and aligned/steady current legs are proven no-gos unified by a single magnitude-homogeneity obstruction; the driven leg is constructively realizable but dynamically un-selected; and the sole regularity gate is closed conditionally by an exact Lamb-vector identity that linearizes the enstrophy inequality, with an explicit, falsifiable  $O(1/\text{Re})$  drive criterion and a clean  $\text{Pm} = 1$  Hall lift. The open frontier is sharply named: prove (3) at the object's Reynolds, and control the  $\text{Pm} \neq 1$  coupled enstrophy.

## References

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*Verification scripts (identity, Gronwall closure, Hall coercivity) and full derivations: [results/verify/](#) in the project repository. No number in this paper is fabricated; ASCII apart from standard math symbols.*